



End Semester Examination (Placement drive /Year Back), August 2026

Roll no.....

Name of the Program and Semester: B. Tech (CSE) AI-ML, 6th Semester

Name of the Course: Transformer Model and Applications

Course Code: TCS-666

Time: 3 hours

Maximum Marks: 100

Note:

- All Questions are compulsory
- Answer any two sub questions among a, b and c in each main question.
- Total marks in each main question are twenty
- Each question carries 10 marks.

Q1.

(10×2=20 Marks) CO1

- Describe masked attention in Transformers. Explain how masking prevents attention to future tokens, maintains causal order, affects gradient flow, and enables autoregressive text generation.
- Explain the role of the Feed-Forward Network (FFN) in a Transformer. Why is it applied independently to each position, and why is it needed after the attention layer?
- What is convergence in Transformer model training? How can you determine whether the model has reached convergence?

Q2.

(10×2=20 Marks) CO2

- Explain the learning rate warmup and decay used in Transformer training. Why is warmup needed, how does it improve training stability, and what happens if warmup is not used?
- Explain different batch processing techniques used in deep learning, including fixed-size batching, dynamic batching, padding, and masking. Also discuss how improper padding can reduce computational efficiency.
- Discuss the concept of Batch Normalization in model training with suitable example.

Q3.

(10×2=20 Marks) CO3

- Describe the text generation process in GPT (Generative Pre-trained Transformer) models. Illustrate the process with a suitable example and code.—
- Discuss how BERT is used for different NLP tasks and compare it with its major variants also describe BERT and its model design.
- Compare Machine Translation and Text Summarization using Transformer-based NLP models



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in terms of architecture, objectives, training data, evaluation methods, and real-world challenges.

Q4.

(10×2=20 Marks) CO4

- a. Explain how a Vision Transformer converts an image into a sequence of tokens and how this helps the Transformer process image data.
- b. Describe the architecture of a Transformer for audio processing and explain its similarities and differences with text-based Transformers.
- c. Discuss cross-modal learning and how it enables models to jointly process and learn from multiple data modalities.

Q5.

(10×2=20 Marks) CO5

- a. Explain how sequence length affects the computational cost of Transformers and why longer sequences create scalability problems.
- b. Explain dataset bias and discuss how biased data can affect the performance and fairness of machine learning models.
- c. Explain the trade-off between computational efficiency and accuracy in efficient Transformer models.